

THUSO NDOU

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PROFESSIONAL SUMMARY

BSc Information Technology graduate from North-West University, currently pursuing a BSc Honours in Data Science at Eduvos. Founder and Software Developer at Knolink with hands-on experience developing full-stack applications, AI systems, data-driven solutions, APIs and cloud applications. Experienced across software development, data analysis, artificial intelligence and cloud computing, with practical experience using Python, Java, TypeScript, React, Node.js, SQL, MongoDB and Microsoft Azure.

SKILLS

- AI & Machine Intelligence: AI Application Development, Microsoft Foundry, Multi-Agent Systems, AI Inference, Reinforcement Learning, OpenAI APIs, Vector Databases, Embedding Pipelines
- Cloud & Infrastructure: Microsoft Azure, Azure App Service, Cloud Computing, Cloud-Native Application Development, Docker, Application Deployment, Linux
- Software Engineering: Python, TypeScript, JavaScript, Java, C#, React, Node.js, Express.js, REST API Development, Software Architecture, Full-Stack Development
- Data & Databases: Data Analysis, Data Modelling, SQL, MongoDB, Database Design & Management
- Security & Systems: Authentication & Access Control, Secure Coding, WebSocket, Server-Sent Events, System Analysis & Design, API Development & Integration
- Tools: Git, GitHub, Tailwind CSS

EDUCATION

Bachelor of Science in Information Technology

North-West University
2022 – 2025 | Completed

BSc Honours in Data Science

Eduvos
2026 – In Progress | Incomplete

WORK EXPERIENCE

Founder & Software Developer - Knolink

Jul 2025 – Present

- Identified a gap in the academic research workflow and developed a product strategy around AI and cloud-based solutions.
- Architected and developed the platform end-to-end, including frontend, backend, database architecture and AI integrations.
- Designed and implemented secure authentication, data validation and cloud deployment using Microsoft Azure.
- Manage the product lifecycle from ideation and development through deployment, user onboarding and continuous improvement.
- Engage directly with students and researchers to gather feedback and translate user requirements into product improvements.
- Research and implement emerging AI and cloud technologies to improve platform capabilities and performance.

Developer - BlueGrid Digitals

Mar 2024 – Jun 2025

- Implemented secure coding practices, including password policies and access controls.
- Developed and maintained web and mobile applications for multiple clients.
- Built and integrated REST APIs, frontend interfaces and database solutions using Python, Java, C# and JavaScript.
- Collaborated with clients to gather requirements and deliver functional, user-focused solutions.
- Debugged and resolved software issues during development and testing.
- Assisted with system configuration and application deployment in development environments.
- Automated internal processes to improve development and operational efficiency.

PROJECTS

Roots Online Butchery Platform - Personal Project

2024

A complete end-to-end e-commerce platform for a butchery business, featuring Stripe payment processing, JWT token-based authentication, real-time inventory management, MongoDB aggregation pipelines and a comprehensive admin dashboard.

Role: Full-Stack Developer. End-to-end platform development, including the React frontend, Node.js/Express REST API, Stripe Checkout webhook integration, role-based access control, MongoDB data modelling.

Key Learning: Payment gateway integration, idempotent order lifecycle tracking, role-based access control, MongoDB indexing and aggregation pipelines, reactive state management with caching and system design.

Skills: React, Node.js, Express, MongoDB, Stripe API, JWT, Tailwind CSS, Git

Client Web & Mobile Application Development - BlueGrid Digitals

2024

Developed and maintained web and mobile applications for various clients, including frontend interfaces, backend APIs and database solutions.

Role: Developer. Built and integrated REST APIs, developed frontend interfaces, collaborated with clients to understand requirements and delivered functional, user-friendly solutions.

Key Learning: Client communication, requirements gathering, full-stack development, debugging and testing, secure coding practices and deployment to development environments.

Skills: Python, Java, C#, JavaScript, Node.js, Git, Linux, Windows

Knolink Machine Intelligence Suite - Personal Project

2025

Unified infrastructure for multi-agent inference, coordinating inference runtimes, AST compiler intelligence, and autonomous reinforcement learning across specialised compute nodes. Production-ready system focused on reliable, scalable, low-latency AI processing.

Role: Founder and Lead Engineer. Architected and built the entire system end-to-end, including the distributed orchestration layer, vector retrieval pipeline with Pinecone and OpenAI embeddings.

Key Learning: Distributed systems architecture, vector database optimisation, embedding pipelines, multi-agent orchestration, production deployment strategies and performance tuning for low latency AI inference.

Skills: Azure App Service, Docker, TypeScript, Pinecone, OpenAI API, Firebase, Tailwind CSS, Node.js, Git, Microsoft Foundry

Apolemia - Knolink

2025

A high-throughput inference execution engine designed for real-time model serving. Features dynamic KV-cache reuse, activation sparsity routing, enclave memory security, and low-latency streaming token distribution.

Role: Lead Developer and Systems Architect. Designed and implemented the inference runtime, including the dynamic KV-cache reuse pipeline, routing logic, resilient WebSocket and SSE token streaming dispatch

Key Learning: LLM inference optimisation, KV-cache management, activation sparsity techniques, real-time streaming protocols, hardware-level security (enclave sandboxing) and high-throughput system design.

Skills: TypeScript, Python, React, Node.js, WebSocket, Server-Sent Events (SSE), Tailwind CSS, Git

Entropy - Knolink

2026

A high-performance developer IDE and AST compiler environment designed for deep code intelligence. Built with modular Rust crates and a reactive TypeScript interface for fast syntax processing and deterministic code editing.

Role: Lead Developer and Compiler Engineer. Designed the modular Rust crate architecture, built the high-speed AST parser and static analyser and developed the deterministic multi-pane editing canvas.

Key Learning: Rust systems programming, AST parsing and compiler design, WebAssembly integration, UI state management, zero-latency syntax highlighting algorithms, code analysis and browser-native performance.

Skills: TypeScript, React, WebAssembly, Tailwind CSS, Git

Praxis - Knolink

2026

An experimental reinforcement learning environment studying how strategic planning emerges from autonomous feedback, deep Monte Carlo Tree Search (MCTS) and value network convergence.

Role: Lead Researcher and Developer. Designed the reinforcement learning environment, implemented MCTS rollouts and self-play mechanisms, and analysed value network convergence behaviour.

Key Learning: Reinforcement learning fundamentals, Monte Carlo Tree Search (MCTS), self-play training methodologies, value network convergence analysis, autonomous agent design and experimental research methods.

Skills: Python, Reinforcement Learning, MCTS, Git

AWARDS & SHORT COURSES

Sep 2022 — Cisco Introduction to Cybersecurity (Cisco)

Mar 2023 — IBM Data Science Foundations- Level1 (IBM)

Apr 2023 — IBM Data Science Tools (IBM)

Dec 2022 — MTN Business App Academy Intermediate-POE (Certificate of completion) (MTN)

Jul 2024 — Ayoba Business Chat API Hackathon (Ayoba)

Dec 2024 — Chongqing Polytechnic University of Electronic Technology Basic Training on Network Security Management (Chongqing Polytechnic University of Electronic Technology)

2023 — Golden Key International Honour Society Member (Golden Key International Honour Society)

PORTFOLIO & LINKS

Knolink — <https://knolink.co.za>

GitHub — <https://github.com/Thuso-Ndou>

Personal Website — <https://virtual-cv-roan.vercel.app/>

Praxis — <https://praxis.knolink.co.za>

Apolemia — <https://apolemia.knolink.co.za>

Entropy — <https://entropy.knolink.co.za>

REFERENCES

- References available upon request